Urban effective-service control: a proposed timestep methodology

Selective-discreteness urban review

27 October 2026

Feedback 10 status. This document preserves an earlier candidate transcription. The user selects shared left/through permission for the starting bay case, actual-admission commitment, unassigned movement-priority residuals and average receiving A. The logical allocator may inspect other movements after a failed allocation; its physical access/interface with the common feeder remains open.

Frozen waiting targets and old-target rationing are unselected conditional material. Q4 is withdrawn. A still has sequential head-clearance and current-head-sharing dependencies. The updated coupled model is not yet defined by the old equations below. This is an answer/status update only; see the separate feedback 10 answer.

1 The traffic question and admitted world

A vacant left-turn bay is useful only if a turner can reach it. A through block in the common feeder can prevent access: additional left service need not help, while through service can remove the obstacle. The question is whether continuous mass with exact local semantic order predicts useful control responses that sound aggregate explanations miss.

Retain the accepted isolated four-leg task, V0–V4, fixed routes and four protected effective-service groups: NS through/right, NS left, EW through/right, EW left. Let

$$g_p \geq 0, \quad \sum_{p=1}^{4} g_p = 1, \qquad \mu_{\ell m}(g) = \min\{C_\ell, s_{\ell m} g_{p(m)}\}. \tag{1}$$

Fractions allocate average service, without defining executable phase intervals. Perturbations preserve the sum and nonnegativity. Saturation s and section capacity C have units veh-eq/s. No phase waveform, microscopic lane change, route choice or network-wide lane expansion is introduced.

V0 uses independent movement-exclusive paths. A through/right service class is order-neutral only if its members face equivalent receiving restrictions, or these are nonbinding. V1 has a shared stop-line lane. V2/V3 have the same finite-bay geometry: V2 is constructed so order does not restrict access; V3 retains access-sensitive order. V4 varies declared geometry, demand and receiving conditions on held-out cases. A V2 label alone does not prove an order-neutral trajectory.

Physically split an approach into upstream aggregate section U, common local feeder F, and parallel downstream lanes b beginning at an internal fork. Bay and through continuation have the same fork-to-stop-line distance in the symmetric core, with separately specified capacity/storage. Path length is $L_U + L_F + L_b$; these are disjoint storage owners. V1 omits the fork; V0 uses independent paths, without a synthetic common FIFO feeder. This internal-bay composition extends neutral M2; it is not implemented behavior.

Permissions are an explicit structural input, not inferred from lane count. The exclusive-target illustration used for the matched closure below is $L \to B$ and $T/R \to T$. It is one candidate, not the selected primary matrix. The expanded decision appendix in the compact report also considers left turns in both bay and shared continuation; feedback 10 selects that configuration for the starting bay scenario. It requires a matching comparator, rather than silently applying the exclusive-target ablation. Both V2 and V3 must share the chosen permissions.

2 State, timing and exact order

Each physical section e owns cumulative counts $N^+_{e,k}, N^-_{e,k}$, occupancy $n_{e,k} = N^+_{e,k} - N^-_{e,k}$ and propagation history. A source owns waiting mass $Q_{o,k}$. Ordered ledgers retain route-table reference, route position, continuous mass and eligibility provenance. Route position advances at the actual junction crossing, not at internal $U/F/b$ crossings.

Historical state qualification. The remaining-target vector described next belongs to the earlier waiting proposal. New unadmitted mass has no such binding vector under feedback 10; realized lane occupancy/order remain fixed.

A canonical homogeneous cohort has mass q, one current movement and route reference/position, one owner and compatible contiguous timing. A remaining target vector $a_b \geq 0$, $\sum_b a_b = q$, is a view of mass still upstream, not another physical queue. One cohort may split among legal lanes without an artificial order between its children. Different cohorts are processed in source order; every realized lane retains that subsequence.

Storage fragments do not define traffic behavior. Before allocation/admission, coalesce exactly compatible adjacent fragments into maximal cohorts. Compatibility requires identical route/position, owner, permissions, affine eligibility and, for assigned cohorts, target proportions. Retain M1's exact affine-line test, without tolerance-based mixing. Different commitments or timing are not identical metadata. This canonicalization prevents an arbitrary storage split changing the homogeneous group seen by allocation. It is a proposed transcription requirement, not a new integer packet unit.

The grid is a movement view, not the order authority. If ledger entry r occupies mass interval I_r and cell c is B_c,

$$M_{cm} = \sum_{r:m(r)=m} |I_r \cap B_c|. \tag{2}$$

Discharge consumes the ledger, not an inferred order from M. Rank is mass, not metres. Grid resolution cannot change the exact-ledger traffic law.

3 Inherited boundary-history propagation

Use $t_k = k\Delta t$, empty prehistory and triangular-FD sections with delays τ^f_e, τ^b_e, capacity C_e and storage $K_e = C_e(\tau^f_e + \tau^b_e)$. Require $\Delta t \leq \min_e(\tau^f_e, \tau^b_e)$; a short-section violation requires reconsidering discretization, not clipping delays. For $\theta = (t - t_h)/\Delta t$,

$$\mathcal{H}^{\pm}_e(t) = (1-\theta)N^{\pm}_{e,h} + \theta N^{\pm}_{e,h+1}, \quad t_h \leq t \leq t_{h+1}. \tag{3}$$

Histories are zero for $t < 0$. Compute pre-update allowances:

$$\begin{aligned} A_{e,k} &= [\mathcal{H}^+_e(t_{k+1} - \tau^f_e) - N^-_{e,k}]_+, \\ S_{e,k} &= \min(C_e\Delta t, A_{e,k}), \qquad (4) \\ R_{e,k} &= \min\{C_e\Delta t, [K_e + \mathcal{H}^-_e(t_{k+1} - \tau^b_e) - N^+_{e,k}]_+\}. \qquad (5) \end{aligned}$$

These are masses per update. The first $S_{e,k}$ resident ledger mass is the sending tranche; the first $A_{e,k}$ is available for logical assignment. This inherits discrete readiness; it does not assert that all the tranche was physically ready at every instant within the step. New admissions cannot receive same-section service in this update.

For actual entrance $b_{e,k}$ and exit $y_{e,k}$,

$$N^+_{e,k+1} = N^+_{e,k} + b_{e,k}, \quad N^-_{e,k+1} = N^-_{e,k} + y_{e,k}, \quad n_{e,k+1} = n_{e,k} + b_{e,k} - y_{e,k}. \tag{6}$$

Today's departure creates no immediate extra receiving room. Each internal transfer appears once with each sign in its two physical owners.

After selecting flows, reconstruct each receiving lane's entrance segments from its boundary inflow and retained child order. For total inflow $b > 0$, a child occupying entrance rank $[r, r+q']$ receives time segment $[t_k + \Delta t\, r/b, t_k + \Delta t(r+q')/b]$, with the receiving free-flow shift. This applies M1-style affine reconstruction to the new section graph. Do not retain internal service-event times as new propagation knots. Cross-input ties on outgoing links are disclosed bookkeeping; this isolated task has no later controlled junction where such a tie could confer an order advantage.

4 Known capability inside allocation

This is the proposed exact completion of feedback 9. Separate feasible entry now from remaining intention. Do not first fix a target without known limits.

For this joined proposal, first new assignment occurs at the fork's eligible front, before physical admission is guaranteed. The older prototype placed assignment at entry to F; that location was not selected in the principle discussion. This proposed location change is explicit: it makes current branch limits available without forecasting travel through F. Existing valid waiting targets are still preserved.

At the fork process canonical cohorts in physical source order. Lane-load estimate h_b is pre-update resident branch mass plus target mass assigned by earlier cohorts processed this update, including pending mass. Later cohorts therefore see earlier assignments. Do not subtract predicted signal discharge to invent service-aware driver knowledge. Compare raw mass between matched legal lanes; other perception/cost models remain outside this draft.

Let s be remaining common sending, u_b remaining receiving on reachable legal branches, and q the available cohort mass. For a fresh cohort:

$$\begin{aligned} V &= \min\{q, s, \sum_b u_b\}, \\ x &= \operatorname*{arg\,min}_{0 \le x_b \le u_b,\ \sum_b x_b = V} \frac{1}{2} \sum_b (h_b + x_b)^2, \\ x_b &= \min\{u_b, [\lambda - h_b]_+\}, \quad \sum_b x_b = V. \end{aligned} \tag{7}$$

The strictly convex objective selects unique x; the threshold itself need not be unique on a fully capped plateau. For $V = 0$ set $x = 0$. This is capped continuous queue filling for one cohort, not a heterogeneous equilibrium.

Unselected after feedback 10. The following waiting-target and old-target formulas are preserved for traceability, not adopted as the current model. Their premise is absent under commitment only at actual admission.

To retain logical assignment of waiting traffic, assign remaining intention only after feasible entry has been selected:

$$z_b = [\nu - h_b - x_b]_+, \quad \sum_b z_b = q - V, \qquad a_b = x_b + z_b. \tag{8}$$

Use only legal eventual targets. No current receiving is consumed by z, and it promises no current physical access. The new target a then stays fixed. This second queue fill is an explicit proposed waiting-intent convention, not a user-approved unconstrained-first allocation law.

For valid old remaining targets a, do not allocate again:

$$\bar{u}_b = \min(u_b, a_b), \quad V = \min\{q, s, \sum_b \bar{u}_b\}, \quad x_b = \min\{\bar{u}_b, [\lambda - h_b]_+\}, \quad z_b = a_b - x_b, \tag{9}$$

with $\sum_b x_b = V$. The same proposed queue criterion selects feasible realization within fixed targets. It does not retarget when capacity changes; nor does it claim feedback 9 selected former Candidate B.

Scope of the old prefix gate. The physical prefix statement below applies to an occupied single common lane. It is not the selected general rule for logical allocation-server processing. Reconciling movement-specific inspection with the physical interface remains open in this answer-only update.

Append x_b to branch b, reduce sending/receiving by actual entry, and set $h \leftarrow h + a$. Count $a = x + z$ once, not $x + a$. The residual $q - V$ and targets z remain with the upstream owner. On a genuine common feeder, any residual older mass closes the physical prefix gate for later cohorts this update. Later logical assignment can continue with $V = 0$; it cannot authorize bypass. Independent paths have independent gates.

Cut an unassigned cohort at the inherited eligibility boundary before creating targets, preserving the not-yet-eligible tail. Under empty loading and assignment at this fork, assigned mass was already eligible here; its residual remains eligible while waiting at the same owner. A sending quota can still restrict its actual entry through V. No extra proportional retargeting of a newly ready part is required. Within a homogeneous admitted prefix, lane amounts split that prefix; one target child is not artificially placed before another.

Why the dependency matters. For $h = (0.2, 0)$, $q = 1$, $u = (0.3, 0.5)$ and $s = 0.6$, $x = (0.2, 0.4)$, $z = (0.2, 0.2)$, $a = (0.4, 0.6)$: 0.4 remains upstream. If $u = (0, 1)$ and $s = 1$ instead, $x = a = (0, 1)$. A target

$(0.4, 0.6)$ frozen without those known limits would admit only 0.6. Conversely, if that vector really is an older valid target, the 0.4 cannot silently be reassigned.

5 Elapsed service and current-head receiving reuse

Expose only each lane's current ordered head in its pre-state sending tranche, with rate cap $c_\ell = \mu_{\ell m}(g)$. A zero-rate head blocks that lane. For the proposed timestep node solver define

$$\rho_j = R_{j,k}/\Delta t. \tag{10}$$

This is an explicit step-average computational convention, not the physical availability curve inside the interval. It spends at most the inherited receiving allowance and writes endpoint history only. Event histories remain deferred.

For outgoing link j, collect current heads from leg i. In this isolated topology each leg-to-output movement has one relevant service group:

$$C_{ij} = \sum_{\ell \in \mathcal{L}_{ij}} c_\ell, \quad w_{ij} = g_{p(i,j)}, \quad V_j = \min(\rho_j, \sum_i C_{ij}),$$

$$r_{ij} = \min(C_{ij}, \theta_j w_{ij}), \quad \sum_i r_{ij} = V_j, \tag{11}$$

$$r_\ell = \min(c_\ell, \eta_{ij}), \quad \sum_{\ell \in \mathcal{L}_{ij}} r_\ell = r_{ij}. \tag{12}$$

Remove zero-cap competitors; $V_j = 0$ gives zero rates. Weight each leg once, with capped reuse; share equally among its lanes subject to individual caps. Independent outputs have independent solves because each lane has one current head/output.

Hold rates until the earliest head/tranche exhaustion or step end:

$$\delta = \min\left\{\Delta t - \sigma, \ \min_{\ell: r_\ell > 0} \frac{\min(q_\ell, S_\ell^{\mathrm{rem}})}{r_\ell}\right\}. \tag{13}$$

Advance every active lane by $r_\ell \delta$, process tied events together, expose new heads and recompute sharing. Stop if all rates vanish. A finite canonical ledger gives finitely many head events; use no mass cutoff or software scan priority. This internal clock calculates one step's output; it creates no new observation grid.

For ready left mass q_L followed by sufficient through mass, with nonbinding receiving/capacity,

$$y_L = q_L, \qquad y_T = s_T g_T \left(\Delta t - \frac{q_L}{s_L g_L}\right), \quad 0 < \frac{q_L}{s_L g_L} < \Delta t. \tag{14}$$

With $\Delta t = s_L = s_T = 1$, $q_L = 0.2$ and $(g_L, g_T) = (0.4, 0.6)$, $y_T = 0.3$. Transferring small ϵ from through to left gives $\mathrm{d}y_T/\mathrm{d}\epsilon = 0.25$: faster head clearance outweighs the smaller through rate on this branch. This hand case is not a complete-TST or usefulness result. Other active constraints can change or nullify it.

6 Complete update and feasibility

Read pre-state histories/ledgers; compute all S/R and logical assignment fronts; process source/internal admissions and forks; solve stop-line service on pre-state tranches; commit paired flows; reconstruct boundary eligibility and grid views; update histories, sources, pending views and sinks. Independent calculations may be evaluated in either software order, but no mechanism spends another's allowance or serves new same-section inflow.

A one-in/one-out internal connection transfers $f = \min(S_{\mathrm{up}}, R_{\mathrm{down}})$ from its resident prefix. Release $D_{o,k}$ joins source waiting at the boundary; admission is bounded by $Q_{o,k} + D_{o,k}$ and first-section receiving, in source order. Independent source paths own separate queues. A sink removes $\min(S, \bar{S}_k)$, where $\bar{S}_k$ is its declared allowance. The central junction uses the current-head law above; there is no hidden additional conflict rule.

The timestep restriction gives $S_{e,k} \le n_{e,k}$ and $R_{e,k} \le K_e - n_{e,k}$. Thus

$$0 \le n_{e,k} + b_{e,k} - y_{e,k} \le K_e, \qquad 0 \le b_{e,k} \le R_{e,k}, \quad 0 \le y_{e,k} \le S_{e,k}. \tag{15}$$

Internal paired flows and route-preserving splits cancel, giving

$$\sum_o Q_{o,k} + \sum_e n_{e,k} + Z_k = D_{<k}, \tag{16}$$

with cumulative sink exit Z_k and exogenous release $D_{<k}$. This holds route by route; pending commitments add no stock. These are definition-level arguments, not tested implementation evidence.

7 One-step derivatives

Work on a fixed order/structural and active-constraint signature. Differentiate continuous split masses, queues, pending targets and event durations, without differentiating route labels or permissions.

For capped queue fill with free set $\mathcal{A}$ and upper-capped set $\mathcal{B}$, if $\mathcal{A} \neq \varnothing$,

$$\begin{aligned} \lambda &= \frac{V - \sum_{\mathcal{B}} u_b + \sum_{\mathcal{A}} h_b}{|\mathcal{A}|}, & \dot\lambda &= \frac{\dot V - \sum_{\mathcal{B}} \dot u_b + \sum_{\mathcal{A}} \dot h_b}{|\mathcal{A}|}, \\ \dot x_b &= \dot\lambda - \dot h_b \ (b \in \mathcal{A}), & \dot x_b &= \dot u_b \ (b \in \mathcal{B}), \quad \dot x_b = 0 \ \text{(zero set)}. \end{aligned} \tag{17}$$

Use $\bar u$ for old targets, retaining $\dot a$ on active target caps. Differentiate V through its active minimum. Fresh waiting intent uses the same free-set calculation with base $h+x$, total $q-V$, no finite upper caps; old intent has $\dot z = \dot a - \dot x$. On fully capped branches $\dot x = \dot u$; on a stable zero-volume branch $\dot x = 0$. Transitions may be kinks, not ordinary derivatives.

For leg sharing with uncapped set $\mathcal{U}$ and capped set $\mathcal{B}$,

$$\dot\theta = \frac{\dot V - \sum_{\mathcal{B}} \dot C_{ij} - \theta \sum_{\mathcal{U}} \dot w_{ij}}{\sum_{\mathcal{U}} w_{ij}}, \qquad \dot r_{ij} = \begin{cases} \dot C_{ij}, & i \in \mathcal{B}, \\ w_{ij}\dot\theta + \theta \dot w_{ij}, & i \in \mathcal{U}. \end{cases} \tag{18}$$

Within-leg sharing uses the unweighted analogue. For an interior head event $\delta = q/r$, $\dot\delta = \dot q/r - q\dot r/r^2$. For total output $y = \sum_a r_a \delta_a$,

$$\dot y = \sum_a (\dot r_a \delta_a + r_a \dot\delta_a). \tag{19}$$

Do not freeze head-clearance times. The final interval's derivative is minus the sum of previous duration derivatives. Ties need compatible branches or explicit one-sided analysis.

Differentiate propagation and conservation through their active branches. Fixed delayed query times have fixed interpolation weights, but the history values retain sensitivity. A cut within an affine eligibility segment $[a, b]$ uses fraction $(t-a)/(b-a)$ and its full derivative. Point segments, zero mass and changes in ledger topology require their boundary branches; the formula is not a global smoothness claim.

8 From discharge to complete TST

Collect all retained numerical state in X_k and write the defined update as $X_{k+1} = F_k(X_k, u_k)$. The inherited left-interval objective is

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} \left[\sum_o (Q_{o,k} + D_{o,k}) + \sum_e n_{e,k} \right], \tag{20}$$

$$P_0 = 0, \quad P_{k+1} = F_{k,X} P_k + F_{k,u} \dot u_k, \quad \dot J = \sum_k (c_{k,X} P_k + c_{k,u} \dot u_k). \tag{21}$$

Within control $\dot u_k = d$; under candidate-independent continuation $\dot u_k = 0$, while P_k keeps propagating. Candidate-independent preparation gives $P_0 = 0$; otherwise differentiate it. No per-vehicle trajectory derivative is needed.

At fixed state the direct partial of this step's cost with respect to service is zero. Changed service can still alter next queues, access, spillback, source waiting and eventual sink exit. Internal transfer, including

a stop-line crossing into another road, does not itself remove system population. For fixed releases, an independent accounting identity is

$$J_{\Delta t} = \Delta t \sum_{k=0}^{H-1} (D_{\le k} - Z_k), \qquad \dot{J}_{\Delta t} = -\Delta t \sum_{k=0}^{H-1} \dot{Z}_k. \tag{22}$$

This counts all affected mass without duplicate pending stock.

Propose a common integer horizon H qualified to clear every compared nearby feasible action, followed by zero occupancy. This chooses the earlier common-horizon option, not a numerical horizon or universal clearance theorem. Finite demand, acyclic routes, positive continuation service and eventually available sinks support clearance. The chosen horizon must still be qualified over the comparison neighborhood. A residual cutoff, nonclearing case or changing integer stop index is not complete-TST differentiability evidence.

9 Fair and reconstructible aggregate comparisons

Allow the original fine record through each rival's declared macro converter or estimator. Disclose states, retained memory, observation times and calibration data. Native subsets are legitimate; successful macro inference counts against needing explicit order. A persistent ordered ledger is meso information even inside a macro-labelled estimator. Separate retrospective replay, observation-updated prediction and forecasting from a common prefix. Each derivative claim also states which observed inputs are held fixed or regenerated under the candidate action. A later observed outcome cannot count as having been predicted before it was observed.

Add one controlled matched aggregate ablation, B_{mix}, alongside source-native B1/B2. It uses the same section graph, S/R, action, releases, loading, continuation and objective, but retains route/movement totals per section without an ordered ledger. Its following closure is defined here, not borrowed wholesale from a native source.

At a common feeder use pre-state composition β_m. In the primary bay geometry each movement has its declared branch:

$$f_m = \beta_m V, \qquad V = \min\left\{ S_F, \min_{m:\beta_m>0} \frac{R_{b(m)}}{\beta_m} \right\}. \tag{23}$$

Ignore zero fractions; add new inflow after removing this mixture. A homogeneous movement with multiple matched legal targets uses the same constrained queue fill, splitting route counts proportionally within that movement. Do not add permissions merely to create lane-choice freedom.

At a mixed stop-line lane set

$$c_\ell = \min\left\{ C_\ell, \left(\sum_{m:\beta_{\ell m}>0} \frac{\beta_{\ell m}}{s_{\ell m} g_{p(m)}} \right)^{-1} \right\}. \tag{24}$$

A positive fraction at zero service makes $c_\ell = 0$; pure lanes reduce to the selective cap. This is homogeneous elapsed work, not recovered order. Set positive priority $w_\ell = \sum_m \beta_{\ell m} g_{p(m)}$ and output fractions $\beta_{\ell j}$. For the primary topology, each leg has at most one competing lane per output. Starting at zero, increase unfrozen r_ℓ proportionally to w_ℓ until a lane cap or output constraint $\sum_\ell \beta_{\ell j} r_\ell \le \rho_j$ binds. Freeze capped lanes and all positively turning competitors of a bound output. Continue the remaining lanes, processing simultaneous ties together, until all are fixed. This defines a deterministic weighted progressive-fill rate law. Consume each lane's pre-state sending quota, recomputing on quota exhaustion. Extending this comparator to several same-leg competitors requires retaining the agreed leg-level sharing hierarchy; no such extension is assumed here.

Neutral equivalence requires matched pure channels, or a single same-service class with equivalent receiving and readiness. Under the declared primary pure-channel cases both laws reduce to the same S/R/service minimum and current receiving priorities. Equal mixed totals alone do not imply equivalence. B_{mix} isolates a declared closure; it does not replace strong native rivals or establish macro impossibility. In particular, its full-FIFO mixture can restrict bay entry more than an unblocked ordered prefix. A V2 failure is then evidence against that aggregate closure, not proof that exact order is necessary. The principal storage-aware rival must first pass the relevant unblocked/storage checks before a V3 advantage can support the intended information claim.

10 Native rivals and independent/external references

B1's working preference is Wei et al.'s link queue model: moving/queued state with native effective green. Its complete causal link/node law, geometry and initialization must be instantiated. The source's short-free-flow-region limitation of its fixed-speed shortcut prevents treating that shortcut as an unconditional spillback model.

B2's working preference is Gong et al.'s lane-exit allocation anchor with a sourced finite-storage/access adaptation. Exit allocation is not finite-bay entry. Liu–Chang supplies a concrete lane-group storage/blockage source; its native signal/timing and overflow variables need reconciliation before a combined B2 is qualified. Native theorems do not automatically transfer. These implementation-dependent choices stay pending under feedback 8; no new class-selection vote is required.

R1 uses independent hand/reference cases: prefix obstruction, partial entry, receiving reuse, rotation, route-mass balance and neutral limits. They are a correctness floor, not external realism.

For R2 declare a static executable program with fixed cycle C^{cyc}, lost intervals ℓ_p, $L = \sum_p \ell_p < C^{\mathrm{cyc}}$, order and offset. Propose

$$G_p = (C^{\mathrm{cyc}} - L)g_p, \qquad s_{\ell m}^{\mathrm{eff}} = s_{\ell m}^{\mathrm{phys}}(C^{\mathrm{cyc}} - L)/C^{\mathrm{cyc}}. \tag{25}$$

Keep these settings, lane behavior and calibration fixed across model directions and candidate actions. The mapping matches nominal average opportunity, not startup loss or realized microscopic discharge.

Use a common finite route population and desired releases on declared common boundaries. Integer reference vehicles are exogenous population, not rounding of internal simulated mass. Every model receives that release list or its disclosed conversion. SUMO's fixed lane behavior determines realized occupancy; do not replay private selective assignments into only one competitor.

External complete time is desired-release-to-arrival time: TripInfo `duration` plus `departDelay`, without adding waiting time. Track the master population so uninserted/unfinished demand cannot vanish from a completed-trip statistic. Clear all demand or report noncompletion. Common loading does not imply equal later states across simulators. Separate state-conditioned replay from common-input prediction. Test feasible perturbation signs/ranking externally before optimization.

11 Status, evidence limits and source boundaries

Current qualification. The summary below describes the earlier candidate. Its frozen-waiting construction is unselected after feedback 10, and the replacement physical access/interface has not been derived in this response.

This proposal joins physical ownership, pre-state S/R, ordered allocation, fixed waiting intent, prefix admission, elapsed service, endpoint history, complete cost and conditional sensitivities. Newly explicit transcription choices are: feasible-entry-first filling with subsequent waiting intention; the same queue criterion within old targets; canonical cohorts; a fork assignment location and explicit permission subfamily; a step-average receiving clock; a qualified common horizon; and B_{mix}. They remain proposed definitions for review, not newly approved decisions.

Numerical geometry/FD parameters, timestep, grid resolution, horizon, native B1/B2 qualification and R2 program values remain later work. No simulator, AD/FD, optimization or SUMO experiment has run in this writing pass. Reference correctness, invariants, neutral equivalence, meso-sensitive forward behavior, derivative correctness and external usefulness need separate evidence. Finite gradients or a falling internal objective cannot establish useful control directions.

- M0 supplies repository LTM/history/TST; M1 supplies the route ledger, prefix reconstruction and exact merge rule; M2 supplies ownership/grid invariants. The approved M0 source basis is Yperman (2007) and UNsim v0.12.0. This pass checks repository transcription, not fresh reproduction of those complete originals.
- The primary links beside B1/B2 identify their sources, not adoption of their complete algorithms. Gong's indexed primary sections establish scope; its complete algorithm is not independently reproduced here. Liu–Chang's journal year is 2011; overflow notation needs verification.
- Official SUMO signal-program and TripInfo definitions. They support interface/accounting definitions, not empirical equivalence.